

BARSTOW-DAGGETT, CA, USA

WMO: 723815

Lat: 34.854N

Lon: 116.786W

Elev: 1917

StdP: 13.71

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23161

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.6	32.5	-1.9	5.3	49.9	1.3	6.3	48.8	34.7	54.7	32.1	55.0	5.5	270	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	108.9	67.9	106.6	66.9	104.4	65.9	72.5	98.5	70.9	97.2	69.4	96.0	10.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	99.5	81.7	62.2	89.9	84.7	58.5	78.5	88.5	37.2	98.8	35.9	96.5	34.5	96.3	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.1	27.7	24.7	DB	23.3	113.4	3.7	2.0	20.6	114.8	18.5	116.0	16.4	117.1	13.7	118.5	
			WB	18.9	76.3	3.3	5.0	16.5	79.9	14.6	82.8	12.7	85.7	10.3	89.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.5	49.7	52.9	59.9	65.6	74.4	84.2	90.0	88.8	81.3	68.8	56.8	48.2	(6)
	DBStd	15.95	5.84	6.47	6.98	7.66	7.86	6.75	4.25	4.06	6.23	7.07	7.24	5.74	(7)
	HDD50	250	77	40	6	1	0	0	0	0	0	0	22	104	(8)
	HDD65	1925	476	339	188	85	16	1	0	0	1	42	256	521	(9)
	CDD50	6992	66	122	313	469	756	1025	1240	1203	939	584	227	48	(10)
	CDD65	3189	0	1	29	104	306	576	775	738	489	161	10	0	(11)
	CDH74	46479	4	51	484	1375	4016	8495	12201	11251	6508	1889	201	4	(12)
	CDH80	28829	0	4	117	554	2175	5485	8282	7492	3879	804	37	0	(13)
(14) Wind	WSAvg	10.8	7.8	9.6	11.6	14.2	14.5	14.0	11.5	10.8	10.2	9.4	8.3	7.7	(14)
(15) Precipitation	PrecAvg	3.9	0.5	0.6	0.5	0.2	0.1	0.1	0.4	0.3	0.5	0.2	0.3	0.4	(15)
	PrecMax	7.5	2.5	2.5	1.9	1.8	0.5	1.3	2.0	1.3	3.3	1.0	1.7	2.0	(16)
	PrecMin	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	1.6	0.6	0.7	0.5	0.4	0.1	0.2	0.4	0.4	0.8	0.2	0.4	0.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.7	79.2	88.3	95.1	103.7	110.9	112.6	110.2	104.8	96.3	83.9	72.7	(19)
		MCWB	52.3	52.9	57.0	60.3	65.4	65.3	69.4	69.9	66.2	62.2	55.2	53.0	(20)
	2%	DB	69.1	75.0	83.1	90.6	99.3	106.9	109.7	107.9	102.0	91.8	79.5	67.5	(21)
		MCWB	49.7	51.7	55.0	58.1	62.9	65.4	68.5	67.9	65.5	60.4	53.7	49.9	(22)
	5%	DB	65.8	70.9	79.4	86.5	95.7	104.3	107.4	105.7	99.6	88.4	75.3	63.9	(23)
		MCWB	47.9	50.1	53.5	56.4	60.6	64.4	67.7	66.5	64.9	58.7	52.4	47.9	(24)
	10%	DB	62.4	66.6	75.6	82.2	91.7	101.3	104.7	103.2	97.0	84.4	71.2	60.3	(25)
		MCWB	46.8	48.4	52.3	54.8	59.6	63.4	66.8	65.6	63.4	56.9	50.7	46.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	56.5	58.9	62.2	69.2	72.3	75.2	74.3	71.5	65.9	58.2	55.7	(27)
		MCDB	67.7	71.6	80.1	90.4	100.8	104.8	102.3	100.0	92.7	85.0	75.7	67.7	(28)
	2%	WB	52.7	53.8	56.6	59.5	64.4	68.6	72.9	72.1	69.8	62.8	55.7	52.9	(29)
		MCDB	63.0	68.8	77.9	86.6	94.5	101.9	98.7	98.2	92.0	85.3	74.7	63.3	(30)
	5%	WB	50.6	51.8	54.8	57.4	61.9	66.2	71.2	70.5	68.3	60.5	53.6	50.4	(31)
		MCDB	60.6	66.4	75.9	82.6	91.5	99.7	97.6	97.0	91.0	83.9	72.2	60.4	(32)
	10%	WB	48.5	50.0	53.2	55.6	60.1	64.2	69.7	69.0	66.3	58.6	51.9	47.9	(33)
		MCDB	59.2	63.9	73.0	79.4	88.7	97.3	96.6	95.8	90.1	80.8	68.9	57.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.5	23.9	26.0	26.7	27.9	30.1	29.2	29.2	28.4	27.2	25.6	22.9	(35)
		MCDBR	27.7	29.6	30.6	31.5	31.7	32.5	31.5	30.5	29.9	30.6	29.6	26.8	(36)
		MCWBR	14.2	13.8	12.5	12.2	11.4	11.0	11.2	11.0	10.9	12.8	14.1	14.5	(37)
	5% WB	MCDBR	22.1	25.2	28.3	29.3	30.5	31.5	27.2	27.0	26.6	27.7	26.9	22.6	(38)
		MCWBR	12.1	12.2	12.0	11.7	12.3	12.7	12.2	12.0	10.7	11.8	13.0	12.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.273	0.282	0.294	0.326	0.336	0.337	0.371	0.351	0.330	0.309	0.275	0.274	(40)
	taud		2.546	2.534	2.492	2.391	2.391	2.405	2.376	2.417	2.456	2.489	2.576	2.565	(41)
	Ebn at Noon		301	309	312	304	300	298	287	292	295	294	298	294	(42)
	Edh at Noon		25	28	32	37	38	37	38	36	33	29	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1028	1343	1814	2244	2523	2700	2462	2306	2008	1544	1152	929	(44)
	RadStd		74	114	73	69	91	96	111	84	62	71	46	58	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-3.01	N/A	-0.97	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (40 neighbors)	+1.12	N/A	-3.29	+1.03	N/A	N/A	N/A	-178	+391	+211	(47)

Nomenclature: See separate page